Decoding the Zodiac with Machine Learning

Ironhack DS ML Bootcamp
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Agenda

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- EDA
- Horoscope Classification
- Trends & Themes
- Prediction
- Next Steps
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Intro

About Daily Horoscopes

For this project, I am using daily horoscopes written for the 12 western zodiac signs.

Daily Horoscopes are a media staple. They're purposefully written to be generic, so you can say, "Yes, today's new moon *did* open my eye to new possibilities".

Sign	Sign Order	Element
Aries	1	Fire
Taurus	2	Earth
Gemini	3	Air
Cancer	4	Water
Leo	5	Fire
Virgo	6	Earth
Libra	7	Air
Scorpio	8	Water
Sagittarius	9	Fire
Capricorn	10	Earth
Aquarius	11	Air
Pisces	12	Water

Data

Data Sources

- 3 kaggle sources of daily horoscopes from
  - horoscope.com (8,784 horoscopes)
  - globe (6,561 horoscopes)
  - The Hindustan Times (1,356 horoscopes)
- All datasets include the sign, date and horoscope
- Sign value counts are balanced in all datasets

Standardisation

- aligned column names
- dropped “category” column from horoscope.com dataset and merged entries
- switched columns to rows in Hindustan Times dataset
- standardised date format
- added sign_id column (1 = aries, 2 = taurus etc)

Data Cleaning

- renamed mis-typed signs
- deleted rows with missing values
- deleted “lucky number” and “lucky colour” content from Hindustan Times
- replaced mentions of the sign in a horoscope with [Sign Name]

Objectives

Objective 1

Using just the content of the daily horoscopes, classify the zodiac sign.

Success: model performance is better than randomly guessing the sign (i.e. 8.3%)

Objective 2

Explore other signals, e.g. trends across season, sentiment, themes.

Objective 3

Generate fake daily horoscopes, inspired by the existing horoscopes.

EDA - TF-IDF

Overall

TF-IDF (ngram_range = (1,2)) is the base I used for the following analyses. All of them show that there is either no clear distinction between the signs' horoscopes, or that it's mostly nonsensical.

Wordcloud

Visual representation of text data where word size represents frequency, in relation to other signs. A bit gimmicky, but easy to understand for humans.

T-SNE

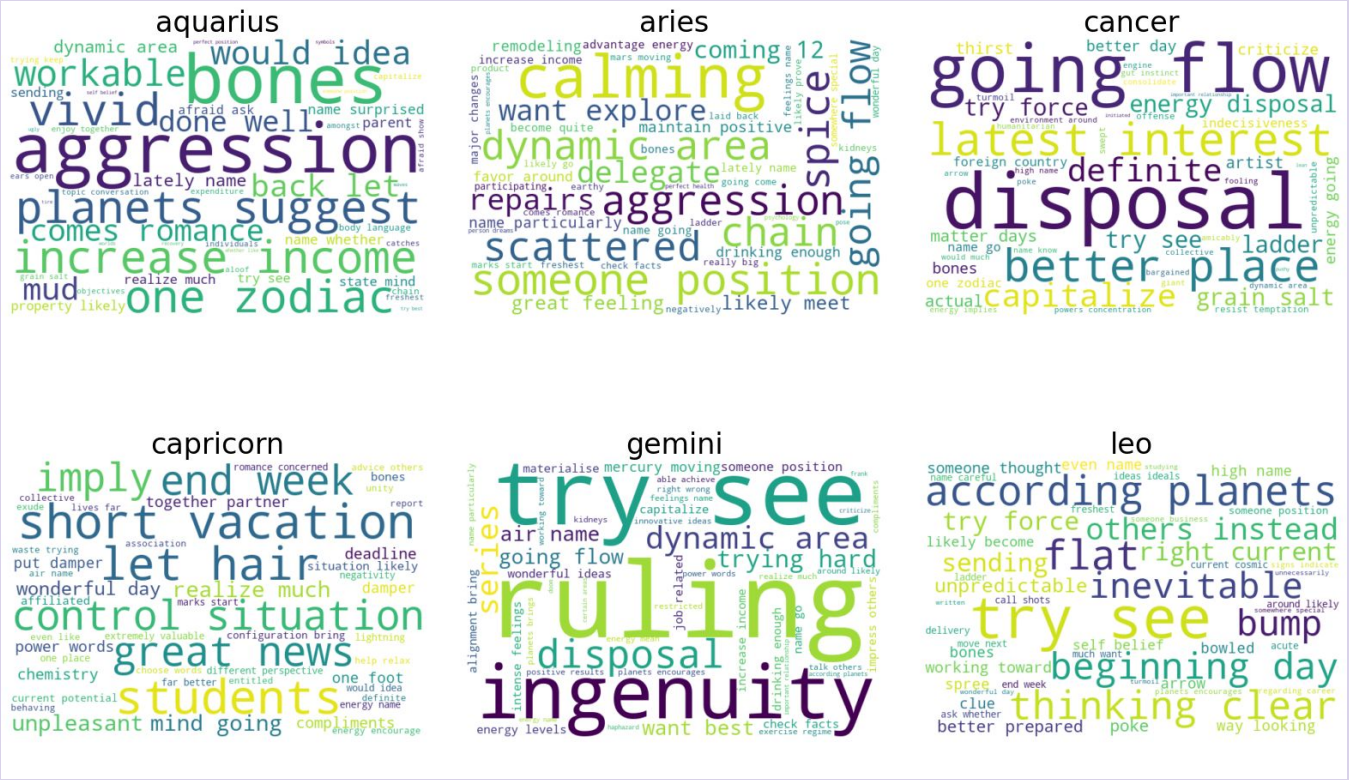
Visually represents high-dimensional data in 2 (or 3) dimensions. Effective at detecting unseen clusters.

Distinctive Words Heatmap

maps top 3 distinctive n-grams per sign on a heatmap

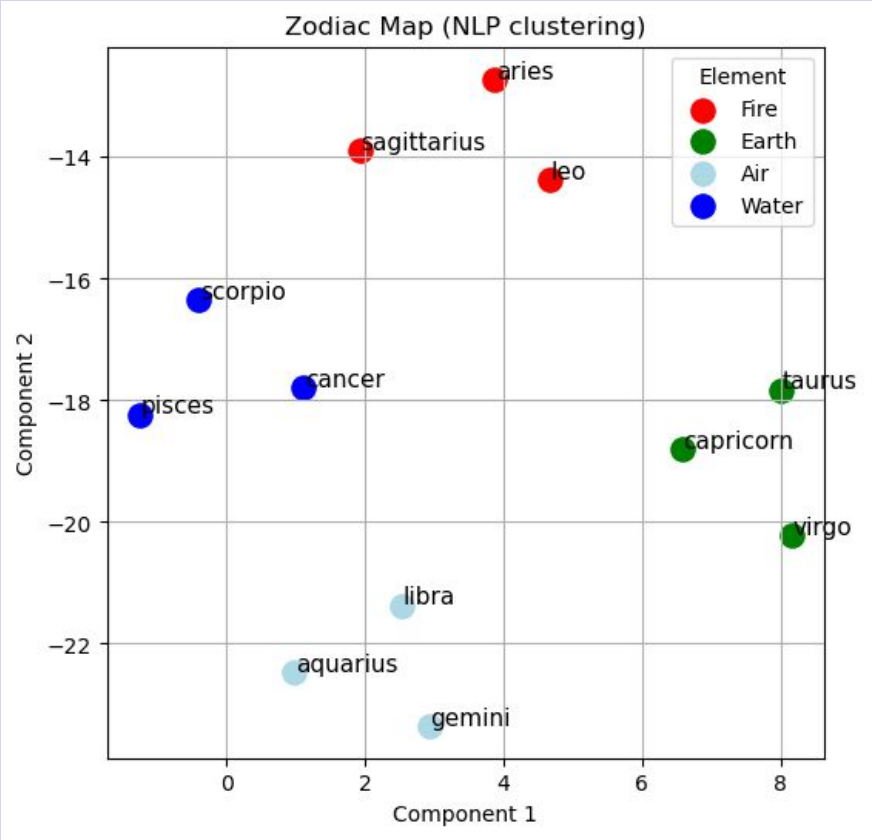
Word Clouds

Experimenting with max_df and custom stopwords gives you very different results.



- custom stopwords: in use
- min_df = 3
- max_df = 0.98

TSNE (T-Distributed Stochastic Neighbor Embedding)



Dimensionality reduction, creates a 2D map based on common features. You can extract and interpret these features.

Component 1:

positive: drinking enough, definite, calming, end week, realize much

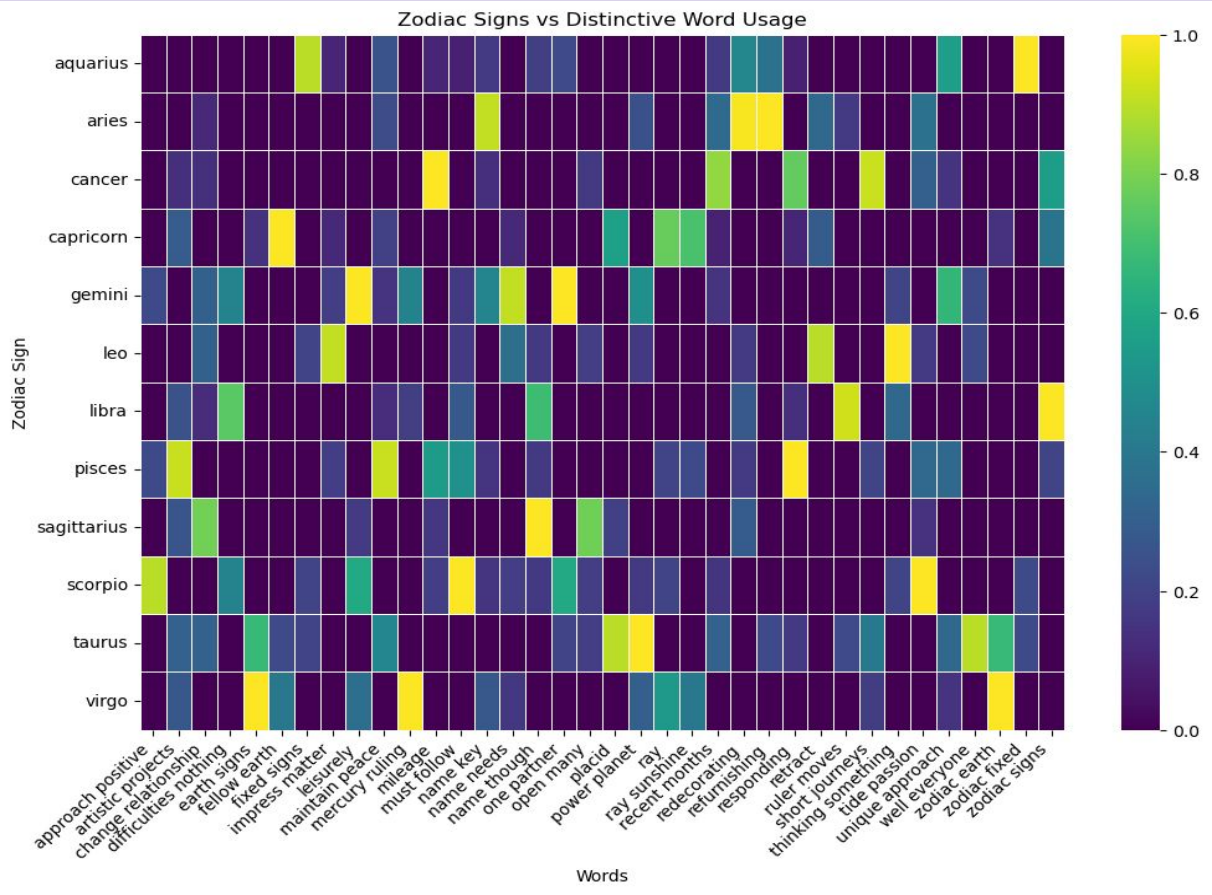
negative: begins retrograde, letter call, sense right, right foods

Component 2:

positive: driver, stray, walk beach, eat eat, throat

negative: bones, arrow, would idea, parent, quiz

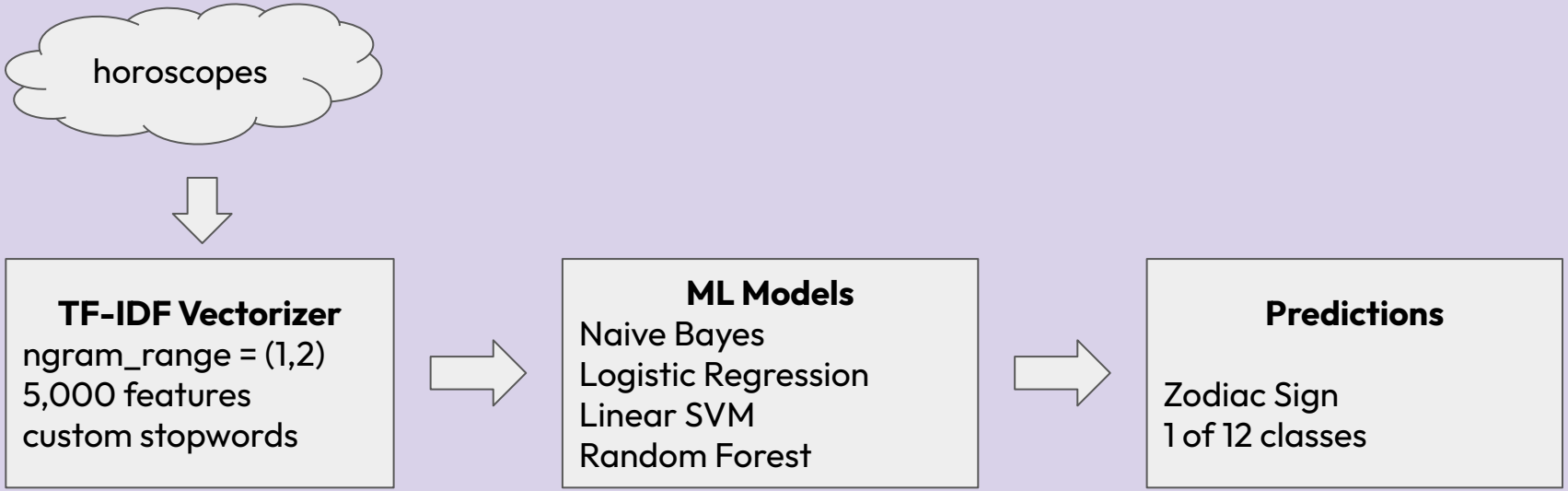
Distinctive Word Usage



Some distinct word pairs per sign emerge, but they're mostly not very insightful.

Horoscope Classification

Pipeline



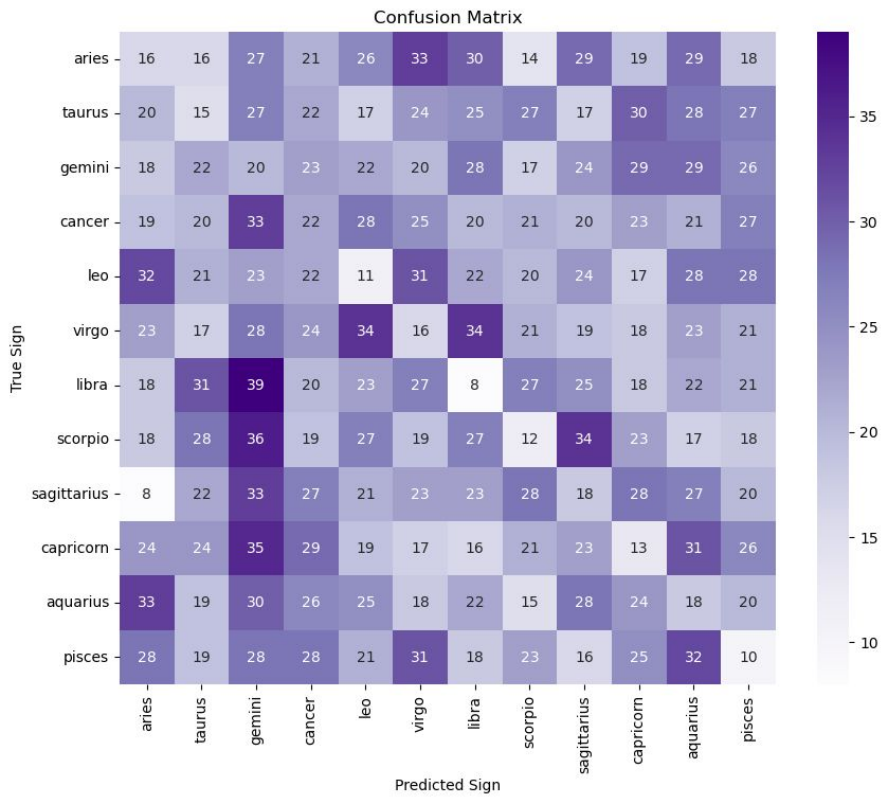
“Best” Model: Linear SVM. Train F1: 0.906, Test F1: 0.057

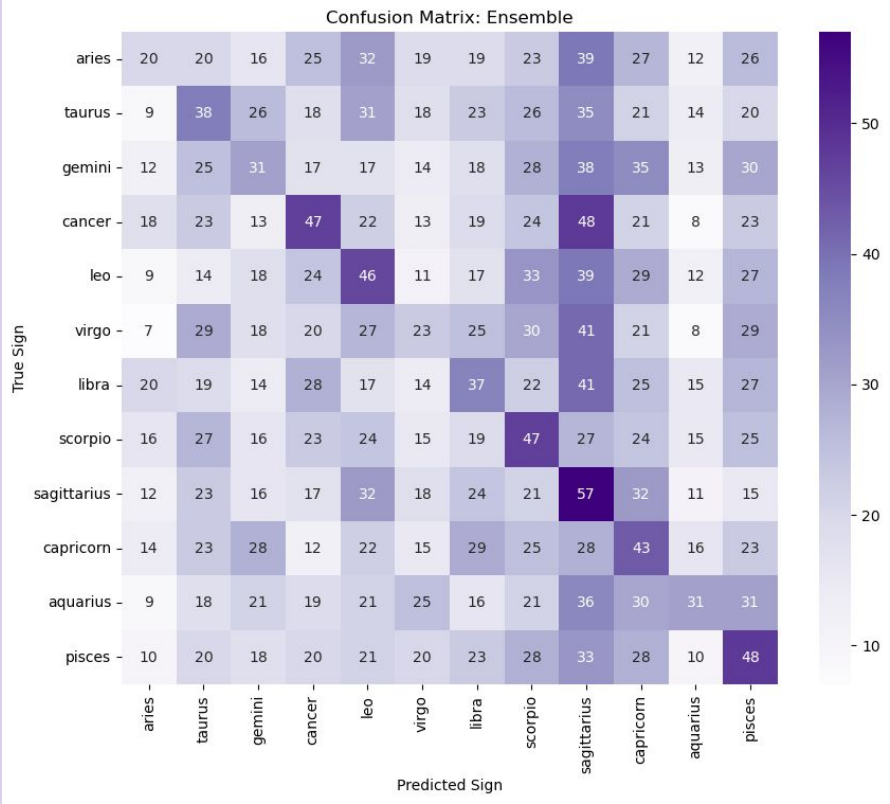
Model Issues and Improvements Tried

All models had huge overfitting issues, while still performing worse than a random guess.

What I tried to solve for	What I did	Effect
Overfitting	Reduced TF-IDF complexity: smaller vocabulary; removed short tokens; adjusted min_df, max_df	Reduced overfitting gap ⭐
Weak signal in short texts	Combined multiple horoscope entries into text chunks to increase context per sample	Worsened model slightly
Models relying too heavily on specific words	Switched to character n-grams (ngram_range =3–5)	Worsened model slightly
Super generic data	Removed custom stopwords - now they're actually a useful signal, and went for aggressive max_df (=0.85) instead	Some model improvement ⭐
Sub-optimal models (RandomForest, Naive Bayes)	Used GridSearchCV on Logistic Regression and Linear SVM, Soft Voting Ensemble	Significant model improvement ⭐

Confusion Matrices





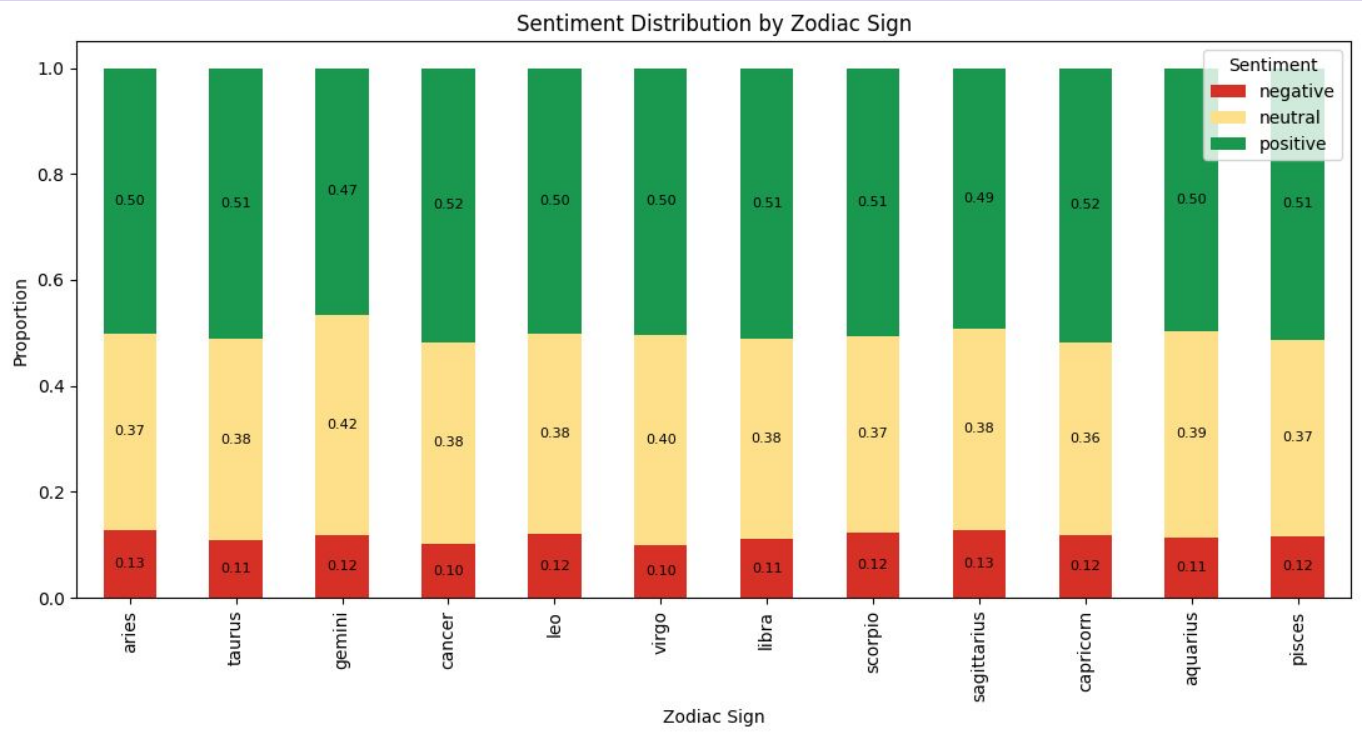
Early Attempt: Logistic Regression
Accuracy: train: 41.6%, test: 6.1%

As good as it's going to get: Ensemble Model
Accuracy: train: 31.2%, test: 14%

Trends & Themes

What patterns emerge in daily horoscopes?

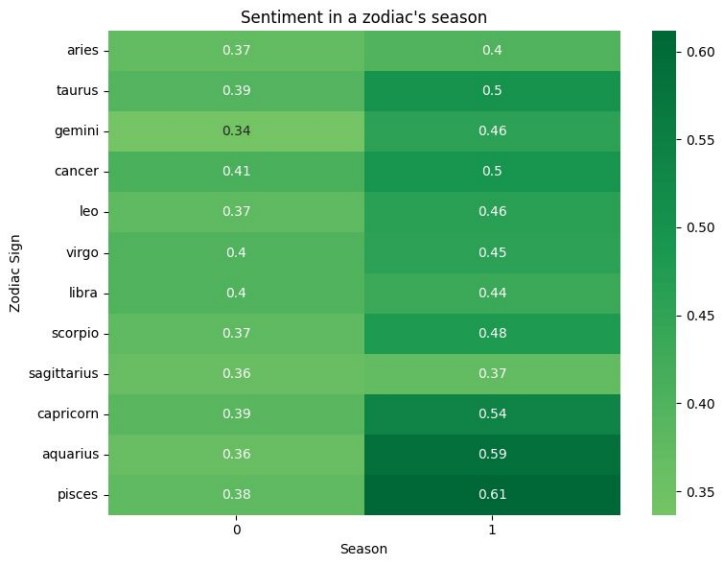
Sentiment Analysis using Transformers - Signs

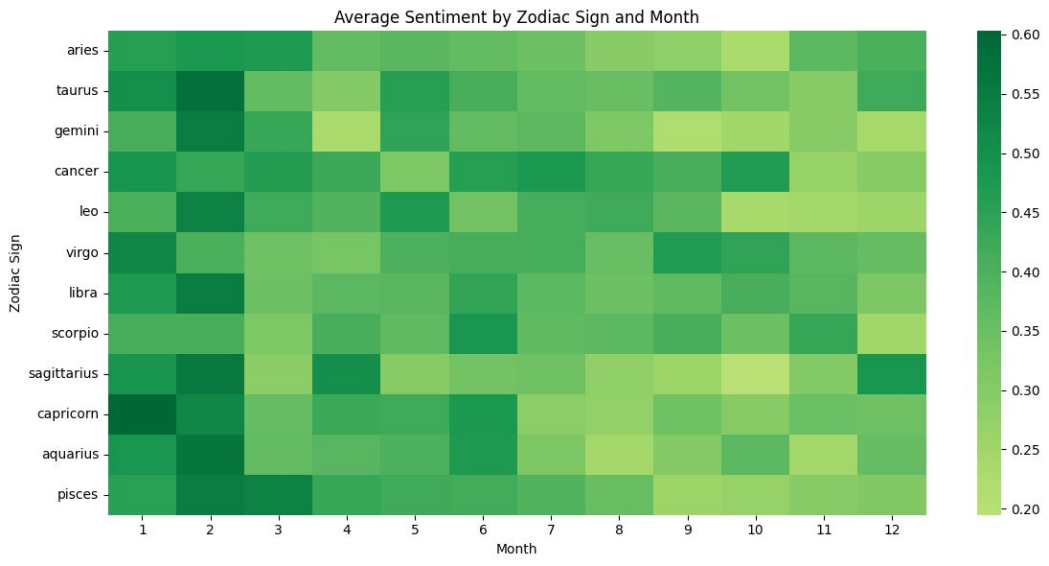


Using a transformer model, I plotted sentiment per sign, to see if some signs had more positive horoscopes than others.

Model: *twitter-roberta-base-sentiment-latest; batch_size = 32, max_length = 512*

Sentiment Analysis using Transformers - Time





Horoscopes are more positive during your birthday season.

Horoscopes are more positive early in the year. (Potentially, since you don't need a bad horoscope to further depress you in February.)

Model: *twitter-roberta-base-sentiment-latest; batch_size = 32, max_length = 512*

Themes

To explore if certain themes were more or less prevalent in different signs/elements, I tried two approaches

Approach 1

- Transformers, using a small zeroshot classification model (142 MB)
- [MoritzLaurer/deberta-v3-xsmall-zeroshot-v1.1-all-33]
- limited text to first 800 characters to improve speed
- It works, but I do not recommend

```
101]  ✓  261m 13.9s
```

Approach 2

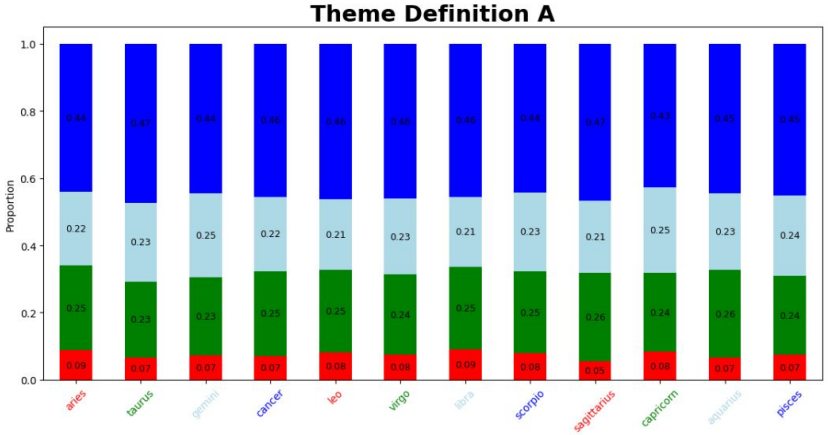
- Sentence Transformers
- [sentence-transformers/all-MiniLM-L6-v2]
- limited text to first 800 characters to improve speed
- Much faster

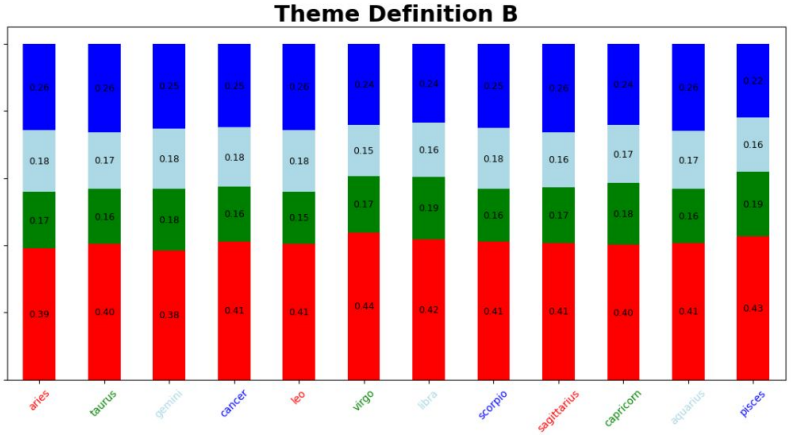
```
[157]  ✓  3m 14.9s
```

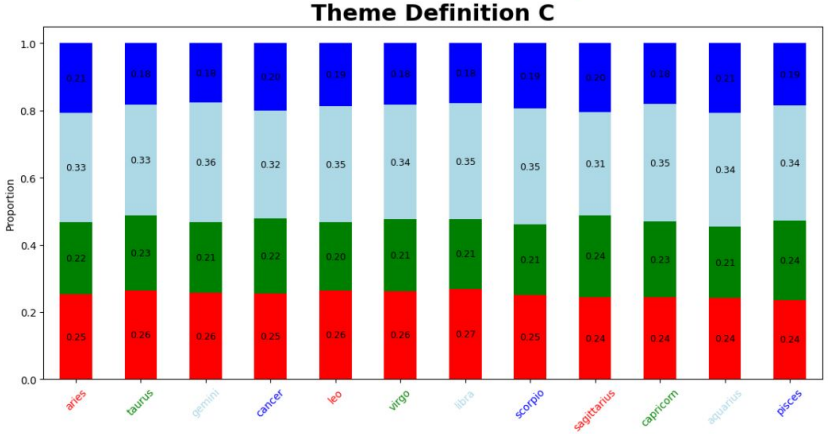
- Using this approach, I could test different theme selections quickly.

Adjusting theme definitions

This clearly shows how you can manipulate outcomes – but the horoscopes remain generic.







Theme Definition A
Fire: action, confidence, leadership, bold decisions
Earth: practical thinking, stability, work and responsibility
Air: communication, ideas, intellectual thinking
Water: emotions, intuition, sensitivity

Theme Definition B
Fire: passionate, action, energy
Earth: loyal, stability, sensual
Air: communication,adaptable,curious
Water: emotion,sensitivity,intuition

Theme Definition C
Fire: bold, confidence, energy
Earth: grounded, reliable, sensual
Air: communication,idea,social
Water: emotion,nurturing,empathy

Horoscope Generation

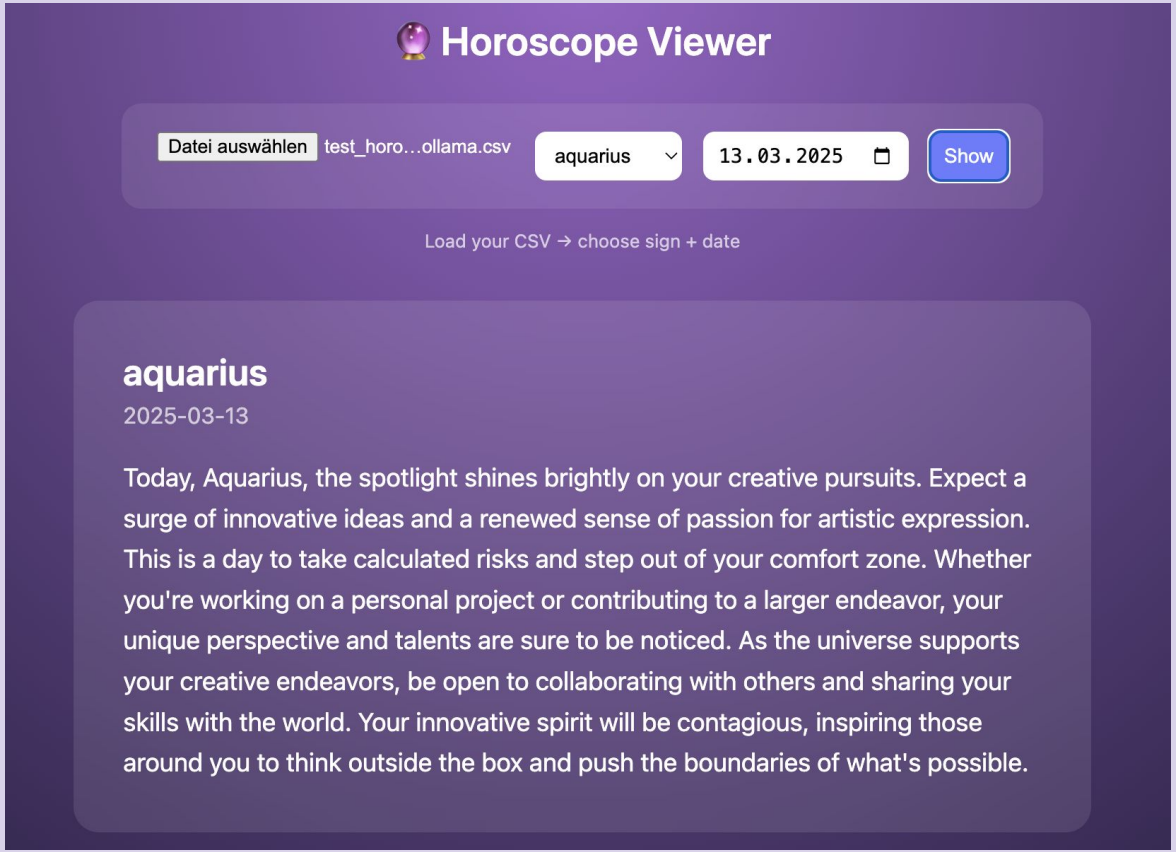
Generating horoscopes using LLaMA 3.2

1. Provide 2 example horoscopes per sign to mimic tone and writing style
2. Prompt the LLM to generate a new daily horoscope
3. Run the generation for every date and every zodiac sign
   a. Requests are processed in parallel to speed up generation
4. Save results periodically to a csv file
5. (Optional if you have a weak computer: Panic because it's Thursday afternoon and you wanted to be done by now.)

```
406m 53.9s

Resuming: 168 already done
 36%|      | 1576/4380 [6:46:41<16:00:21, 20.55s/it]
```

Results: HTML Horoscope Picker



Next Steps & Learnings

Next Steps & Conclusion

This project can be taken further in multiple ways:

1. Classification: Using Transformer models instead, e.g. BERT-based models
2. Deeper analysis on themes and days, e.g. themes on week ends vs week days
3. Horoscope generation: Generating horoscopes inspired by sign characteristics

Conclusion: Daily Horoscopes are mainly fun. They're incredibly generic, on purpose. We can manipulate model inputs to impact output to a certain degree, but even then, horoscope language remains broadly similar across signs.

Any insights we do get, tell us more about humans than the stars

Appendix

Appendix

Github Repo

Kaggle Datasources:

- adxie12/horoscopes
  - globe_horoscopes_scraped.csv
  - horoscopes_scraped_2025
- prasad22/daily-horoscope-dataset
- shahp7575/horoscopes